Name-Vedant Wagh

Roll no.- C 47

```
#include <stdio.h>
#include <stdlib.h>

#define ROWS 10
#define SEATS 7

// Structure for a seat node
struct Node
{
   int seatNo;
   struct Node *prev;
   struct Node *next;
};

// Array of head pointers for 10 rows
struct Node *head[ROWS];

/* Function to create a new seat node */
struct Node *createNode(int seatNo)
{
   struct Node *newNode;

   newNode = (struct Node *)malloc(sizeof(struct Node));

   newNode->seatNo = seatNo;
   newNode->prev = NULL;
   newNode->next = NULL;
```

```
    return newNode;
}

/* Initialize all seats as available */
void initializeSeats()
{
    int i, j;

    for (i = 0; i < ROWS; i++)
    {
        head[i] = NULL;

        for (j = 1; j <= SEATS; j++)
        {
            struct Node *newNode = createNode(j);

            // Insert at the end of doubly linked list
            if (head[i] == NULL)
            {
                head[i] = newNode;
            }
            else
            {
                struct Node *temp = head[i];

                while (temp->next != NULL)
                {
                    temp = temp->next;
                }

                temp->next = newNode;
```

```
            newNode->prev = temp;
        }
    }
  }
}

void displayAvailableSeats()
{
  int i;

  printf("\n--- Available Seats ---\n");

  for (i = 0; i < ROWS; i++)
  {
    struct Node *temp = head[i];

    printf("Row %d: ", i + 1);

    if (temp == NULL)
    {
      printf("No seats available");
    }

    while (temp != NULL)
    {
      printf("%d ", temp->seatNo);
      temp = temp->next;
    }

    printf("\n");
  }
```

```
}

void bookSeat()
{
    int row, seat;
    struct Node *temp;

    printf("\nEnter Row Number (1-10): ");
    scanf("%d", &row);

    printf("Enter Seat Number (1-7): ");
    scanf("%d", &seat);

    if (row < 1 || row > ROWS || seat < 1 || seat > SEATS)
    {
        printf("Invalid row or seat number!\n");
        return;
    }

    temp = head[row - 1];
    while (temp != NULL && temp->seatNo != seat)
    {
        temp = temp->next;
    }

    if (temp == NULL)
    {
        printf("Seat is already booked!\n");
        return;
    }
    // Case 1: Only one node
```

```
    if (temp->prev == NULL && temp->next == NULL)
    {
        head[row - 1] = NULL;
    }

    else if (temp->prev == NULL)
    {
        head[row - 1] = temp->next;
        temp->next->prev = NULL;
    }
    else if (temp->next == NULL)
    {
        temp->prev->next = NULL;
    }

    else
    {
        temp->prev->next = temp->next;
        temp->next->prev = temp->prev;
    }

    free(temp);

    printf("Seat booked successfully!\n");
}

void cancelBooking()
{
    int row, seat;
    struct Node *newNode;
    struct Node *temp;
```

```
printf("\nEnter Row Number (1-10): ");
scanf("%d", &row);

printf("Enter Seat Number (1-7): ");
scanf("%d", &seat);

if (row < 1 || row > ROWS || seat < 1 || seat > SEATS)
{
   printf("Invalid row or seat number!\n");
   return;
}
temp = head[row - 1];

while (temp != NULL)
{
   if (temp->seatNo == seat)
   {
      printf("Seat is already available. Cannot cancel booking!\n");
      return;
   }

   temp = temp->next;
}
newNode = createNode(seat);
if (head[row - 1] == NULL)
{
   head[row - 1] = newNode;
}
else if (seat < head[row - 1]->seatNo)
{
```

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        newNode->next = head[row - 1];
        head[row - 1]->prev = newNode;
        head[row - 1] = newNode;
    }

    else
    {
        temp = head[row - 1];

        // Find correct position
        while (temp->next != NULL &&
               temp->next->seatNo < seat)
        {
            temp = temp->next;
        }

        newNode->next = temp->next;
        newNode->prev = temp;

        if (temp->next != NULL)
        {
            temp->next->prev = newNode;
        }

        temp->next = newNode;
    }

    printf("Booking cancelled successfully!\n");
}

/* Main Function */
```

```
int main()
{
    int choice;
    initializeSeats();
    printf("Welcome to Cinemax Theater Booking System\n");

    do
    {
        printf("\n--- MENU ---\n");
        printf("1. Display Available Seats\n");
        printf("2. Book a Seat\n");
        printf("3. Cancel Booking\n");
        printf("4. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                displayAvailableSeats();
                break;

            case 2:
                bookSeat();
                break;

            case 3:
                cancelBooking();
                break;
```

```
            case 4:
                printf("Thank you for using Cinemax Booking System!\n");
                break;

            default:
                printf("Invalid choice!\n");
        }

    } while (choice != 4);

    return 0;
}
```

Welcome to Cinemax Theater Booking System

--- MENU ---

1. Display Available Seats

2. Book a Seat

3. Cancel Booking

4. Exit

Enter your choice: 1

--- Available Seats ---

Row 1: 1 2 3 4 5 6 7

Row 2: 1 2 3 4 5 6 7

Row 3: 1 2 3 4 5 6 7

Row 4: 1 2 3 4 5 6 7

Row 5: 1 2 3 4 5 6 7

Row 6: 1 2 3 4 5 6 7

Row 7: 1 2 3 4 5 6 7

Row 8: 1 2 3 4 5 6 7

Row 9: 1 2 3 4 5 6 7

Row 10: 1 2 3 4 5 6 7

--- MENU ---

1. Display Available Seats

2. Book a Seat

3. Cancel Booking

4. Exit

Enter your choice: 2

Enter Row Number (1-10): 3